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Chapter 1

Chapter 2

Question 1

Find all rows in “**airquality**” data set that have missing values.

Code

```
rows_missing_val <-airquality[rowSums(is.na(airquality)) > 0, ]
```

Answer

	Ozone	Solar.R	Wind	Temp	Month	Day
5	NA	NA	14.3	56	5	5
6	28	NA	14.9	66	5	6
10	NA	194	8.6	69	5	10
11	7	NA	6.9	74	5	11
25	NA	66	16.6	57	5	25
26	NA	266	14.9	58	5	26
27	NA	NA	8.0	57	5	27
32	NA	286	8.6	78	6	1
33	NA	287	9.7	74	6	2
34	NA	242	16.1	67	6	3
35	NA	186	9.2	84	6	4
36	NA	220	8.6	85	6	5
37	NA	264	14.3	79	6	6
39	NA	273	6.9	87	6	8
42	NA	259	10.9	93	6	11
43	NA	250	9.2	92	6	12
45	NA	332	13.8	80	6	14
46	NA	322	11.5	79	6	15
52	NA	150	6.3	77	6	21
53	NA	59	1.7	76	6	22
54	NA	91	4.6	76	6	23
55	NA	250	6.3	76	6	24

56	NA	135	8.0	75	6	25
57	NA	127	8.0	78	6	26
58	NA	47	10.3	73	6	27
59	NA	98	11.5	80	6	28
60	NA	31	14.9	77	6	29
61	NA	138	8.0	83	6	30
65	NA	101	10.9	84	7	4
72	NA	139	8.6	82	7	11
75	NA	291	14.9	91	7	14
83	NA	258	9.7	81	7	22
84	NA	295	11.5	82	7	23
96	78	NA	6.9	86	8	4
97	35	NA	7.4	85	8	5
98	66	NA	4.6	87	8	6
102	NA	222	8.6	92	8	10
103	NA	137	11.5	86	8	11
107	NA	64	11.5	79	8	15
115	NA	255	12.6	75	8	23
119	NA	153	5.7	88	8	27
150	NA	145	13.2	77	9	27

Chapter 3

Question 2

Find mean, standard deviation, minimum, maximum for each of temperature and ozone level.

Data Manipulation

Extracts the rows in the data without missing values

```
valid_rows <- airquality[rowSums(is.na(airquality)) == 0, ]
```

3.1 Mean

Code

```
mean_temp <- mean(valid_rows$Temp)
mean_ozone <- mean(valid_rows$Ozone)
```

Answer: Mean Temperature:

```
[1] 77.79279
```

Answer: Mean Ozone Level:

```
[1] 42.0991
```

3.2 Standard Deviation

Code

```
sd_temp <- sd(valid_rows$Temp)
sd_ozone <- sd(valid_rows$Ozone)
```

Answer: Standard Deviation of Temperature:

```
[1] 9.529969
```

Answer: Standard Deviation of Ozone Level:

```
[1] 33.27597
```

3.3 Minimum

Code

```
min_temp <- min(valid_rows$Temp)
min_ozone <- min(valid_rows$Ozone)
```

Answer: Minimum Temperature:

```
[1] 57
```

Answer: Minimum Ozone Level:

```
[1] 1
```

3.4 Maximum

Code

```
max_temp <- max(valid_rows$Temp)
max_ozone <- max(valid_rows$Ozone)
```

Answer: Maximum Temperature:

```
[1] 97
```

Answer: Maximum Ozone Level:

```
[1] 168
```


Chapter 4

Question 3

##Calculate Beta estimate

Here, Y is the response variable, and X is the design matrix. The cars data (an R data set, also always available in R) contains two variables: speed and distance to stop. Fit a simple linear regression model to these data, i.e. find the beta estimates, using the equation above, and matrix calculations in R.

Code

```
##### Create design matrix #####
Y <-cars$dist

intercept<- rep(1,50)
X <- cbind(intercept,cars$speed)

##### Calculation #####
beta_hat <- solve(t(X) %*% X) %*% t(X)%*% Y
```

Answer Beta estimate

```
[1] -17.57909
```


Chapter 5

Question 4

##Confirm Beta estimate by fitting model

Check the same estimates as when fitting the linear regression model using `lm()` in R.

Code

```
##### Create design matrix #####
Y <- cars$dist

intercept<- rep(1,50)

X <- cbind(intercept,cars$speed)

##### Fit Model #####
model <- lm(Y ~ cars$speed , data = cars)
```

Answer

Call:

```
lm(formula = Y ~ cars$speed, data = cars)
```

Coefficients:

```
(Intercept)  cars$speed
    -17.579      3.932
```

